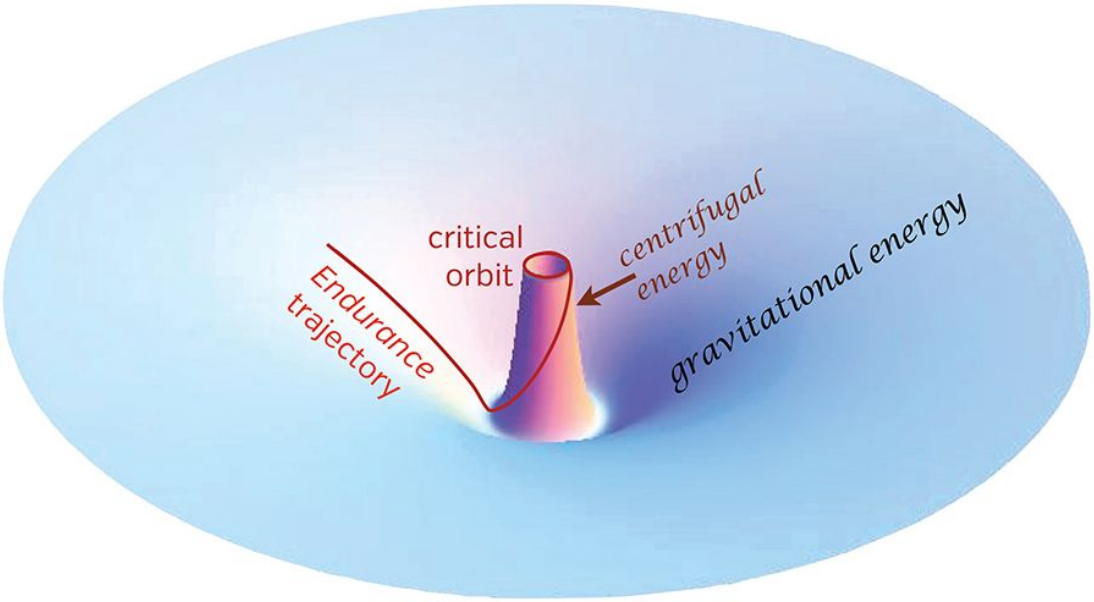


Fig. 27.3. The *Endurance*'s trajectory on a volcano-like surface that represents its gravitational and centrifugal energies.

Think of the surface in Figure 27.3 as that of a smooth, granite sculpture sitting on the floor in your home. It sinks down to a deep moat that surrounds a sculpted volcano.

The *Endurance*, after being pried away from Mann's planet, is like a tiny marble that rolls freely on this granite surface. As it rolls inward toward the moat, the marble picks up speed, because of the surface's downward slope. It then rolls up the volcano's side, slowing as it goes, and arrives on the volcano's rim with some residual circumferential motion. And it then rolls around and around on the rim, delicately and unstably balanced between falling inward, into the volcano, and falling back outward and down to the moat.

The volcano's interior is Gargantua, and the volcano's rim is the critical orbit, from which the *Endurance* launches toward Edmunds' planet.